

Fundamentals of Biology

Science

May, 2026

Fundamentals of Biology (A07954)

Short Title: Fundamentals of Biology
Faculty Science and Computing
Department Science
Cluster Biology
Author AHEARNE
Credits 10

Level: Introductory

Description of Module / Aims

This module will introduce the student to biology, biochemistry and microbiology, focusing on biological terminology and microbiological techniques. It will also introduce the student to the concepts of environmental biology and pollution, and their impact on industry as well as the environment.

Programmes

BIOL -0013 Higher Certificate in Science in Good Manufacturing Practice and Technology 1 / 2 / M
(WD_SGMPT_C)

Indicative Content

- The cell: cell structure; prokaryotic versus eukaryotic and animal versus plant, cell organelles and their roles, fluid mosaic model of biological membranes etc.
- Biological molecules: protein, carbohydrate and lipid structure and biological function
- Enzymology: definition, enzyme properties, enzyme classes and enzyme specificity in biological organisms; factors affecting enzyme activity; applications of enzymes
- Immunology: antibodies, antigens and the immune response; vaccination; applications of antibody:antigen interactions
- Microbiology: introduction to bacteria, fungi and viruses; positive and negative roles of microorganisms; sources of biological contamination, hygiene control; microbiological techniques; biological control
- Environmental biology: environmental biocycles; carbon, nitrogen biocycles; water, soil and air pollution control and monitoring

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use standard laboratory techniques to isolate, purify and stain bacteria and fungi for microscopic examination.
2. Use standard laboratory techniques and rapid detection methods to assess biological contamination, determine hygiene levels and identify a number of micro-organisms.
3. Describe the structure of a number of microorganisms.
4. Describe the concepts of cells, biological molecules, microorganisms and biological contamination.
5. Identify procedures to prevent biological contamination of product and environment.
6. Discuss environmental bio-cycles and the importance of environmental pollution control.
7. Record results of laboratory work using a combination of formal written laboratory reports and worksheets.

Learning and Teaching Methods

- Lectures.

- Laboratory-based practical classes.
- In-class discussions.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	3, 4, 5, 6
Continuous Assessment	50%	
<i>Practical</i>	50%	1,2, 7

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.
- 70%-100%:** All of the above to an excellent level. Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		246

Essential Material

- Campbell, N.A., J.B. Reece, L.A. Urry, M.L. Cain, S.A. Wasserman and Minorsky, P.V., Jackson, R.B. *Biology*. 8th Edition. San Francisco: Pearson, Benjamin Cummings, 2008.

Supplementary Material

- "Moodle ." www.moodle.wit.ie
- Tortora, G.J., B.R. Funke and C.L. Case. *Microbiology - An Introduction*. 10th Edition. San Francisco: Pearson, 2010.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab